

REVIT CONCRETE MODELING

20 Sections in this course outline

20
SECTIONS

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01

Brief overview of Revit (what it entails)

- ✓ Difference with AutoCAD software
- ✓ Benefits over AutoCAD software

02

Setting up the software

- ✓ Brief introduction of Libraries required
- ✓ Downloading other extensions from Autodesk site

03

Beginning with Revit

- ✓ Creating a new project
- ✓ An overview of the templates
- ✓ An overview of palettes, tabs and panels

04

Creating Datum Elements

- ✓ Creating Levels
- ✓ Creating grids – (same as with the levels creation)
- ✓ Propagate options extents on the grid
- ✓ Customizing and changing the shape and appearance of grid bubbles
- ✓ Linking CAD files to create grids

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Brief Overview of Creating 3D elements

- ✓ Foundations, Columns, Beams, Slabs

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Creating Columns

- ✓ Placement of columns
- ✓ Loading column families



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Creating Foundations

- ✓ Hosting foundations on the columns created
- ✓ Creating Strip footing
- ✓ Using eccentricity option
- ✓ Basement floor slab
- ✓ Loading foundation families into your drawings

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Curved Beams

- ✓ Procedure of modeling the curved beams

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Creating Beams and slabs

- ✓ Copying beams and slabs from level to level
- ✓ Creating a slab
- ✓ Hollow pot and waffle slab

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Creating Staircase and Ramps

- ✓ Types of staircase
- ✓ Creating an architectural ramp and structural ramp
- ✓ Modeling Staircase as a ramp
- ✓ Shaft openings

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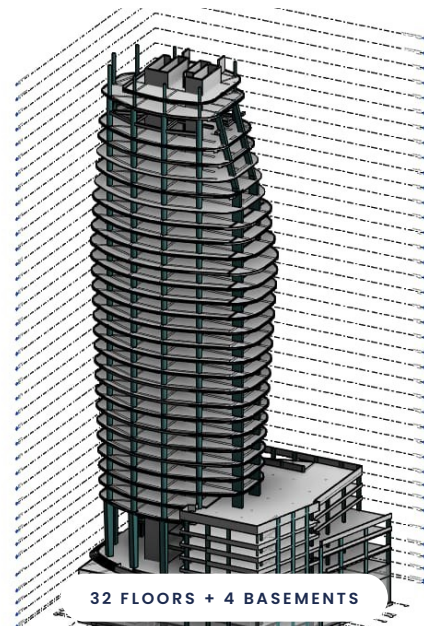
Beam System Layout and Justification Rules

- ✓ Using the beam system to model the structural beams
- ✓ Modelling the beams using the beam system layout rules
- ✓ Modelling the beams using the beam system Justification Rules
- ✓ Deselecting the beam system

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Visibility/Graphics and Filters

- ✓ Applying Visibility/Graphic Overrides — Method 1
- ✓ Applying Visibility/Graphic Overrides — Method 2
- ✓ Applying Visibility/Graphic Overrides — Method 3
- ✓ Applying Visibility/Graphic Overrides — Method 4





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Creating Roofs and Truss placement

- ✓ Sloped and gable ended roofing
- ✓ Adding system trusses on your roof
- ✓ Editing and creating your own customized trusses

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Annotations

- ✓ Adding dimensions
- ✓ Adding spot elevations
- ✓ Adding spot coordinates
- ✓ Adding spot slope

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Schedules and Material Takeoffs

- ✓ Creating Bar Bending Schedules
- ✓ Loading shape images into bar bending schedules
- ✓ Creating Material Takeoffs for the structural elements
- ✓ Sorting and filtering fields in material takeoffs and bar bending schedules

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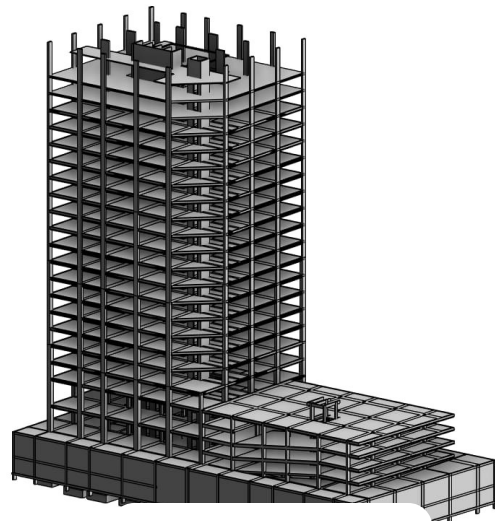
Placing of Reinforcement

- ✓ Creating sections in your drawing
- ✓ Foundation Reinforcement
- ✓ Column Reinforcement
- ✓ Beam Reinforcement
- ✓ Slab reinforcement
- ✓ Staircase reinforcement
- ✓ Ramp reinforcement

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Detailing of Sections

- ✓ Placing breaklines in the sections
- ✓ Using hatch files to create regions
- ✓ Placement of rebar tags on your reinforcements
- ✓ Adding comments on your reinforcement
- ✓ Tagging of the structural elements



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Placing Drawings on Sheets

- ✓ Loading different paper sizes for your title blocks
- ✓ Using system loaded title blocks
- ✓ Editing and customizing the title blocks
- ✓ Adjusting texts and fields in your title block
- ✓ Saving title blocks as a Revit family template
- ✓ Introduction to duplicate options for drawings
- ✓ Placing drawings on the sheets

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Printing Sheets to PDF format

- ✓ Setting up your sheets before printing
- ✓ Printing one sheet at a time
- ✓ Printing several sheets at a time

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Conversion of your files into CAD and other formats

- ✓ Export options for your drawing into other formats
- ✓ Transporting your drawing to AutoCAD
- ✓ Sending your model for analysis

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TOPICS

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Revit Concrete Structure Training

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